

TOPIC :- COMPLETE LIFE CYCLE OF THE CLINICAL TRIAL DATA

Submitted by: Suraj Anil Patil

Guided by: Mrs. Nidhi Jha

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INTRODUCTION

Clinical trials generate a large volume of the data or patients, which need to be accurate and complete for safety and to yield reliable results.

Clinical data management offers a complete lifecycle for data from pre-planning to final analysis in conformity with GCP and regulatory guidelines.

SECTIONS ON THE COMPLETE LIFECYCLE OF CTD

1) PRE-DATA COLLECTION

The clinical data management team reviews the protocol to identify endpoints and key variables that need to be captured. The data management plan details how the data will be collected, validated, cleaned, coded, and locked to meet regulatory guidelines. After case report forms are created that define what needs to be captured, use standard terms, and are devoid of ambiguity, the EDC system (Medidata Rave, Castor, Veeva, etc.) is conformed to the CRFs.

2) DATA COLLECTION

The information is obtained at the time of clinical notes, lab results, hospital stays, and worksheets. Then, the data are collected during the study by directly entering them into the EDC/eCRF at the time of the visit.

According to GCP and ALCOA+, data must be accurate and complete.

3) DATA ENTRY INTO EDC SYSTEM / eCRF

The EDC site coordinators enter the patient's information into the eCRF; for an EDC, this is usually done per visit and per subject.

Data entry is the process of entering the clinical trial data from the source documents (clinical notes, lab results, worksheets, etc.) into the electronic case report form (eCRF) in an electronic data capture system (EDC) such as Castor, Medidata Rave, or REDCap. For most studies nowadays, the sites enter the data into the EDC via the internet and record the subject's data for a particular visit (demographics, VS, labs, AE, study medications).

4) DATA CLEANING AND AUTO-VALIDATION CHECK

The data cleaning process is usually iterative and involves resolving discrepancies and missing data in cooperation with the site. The incomplete or erroneous data can be detected with the help of AI, thus improving accuracy and reducing the time spent on data cleaning.

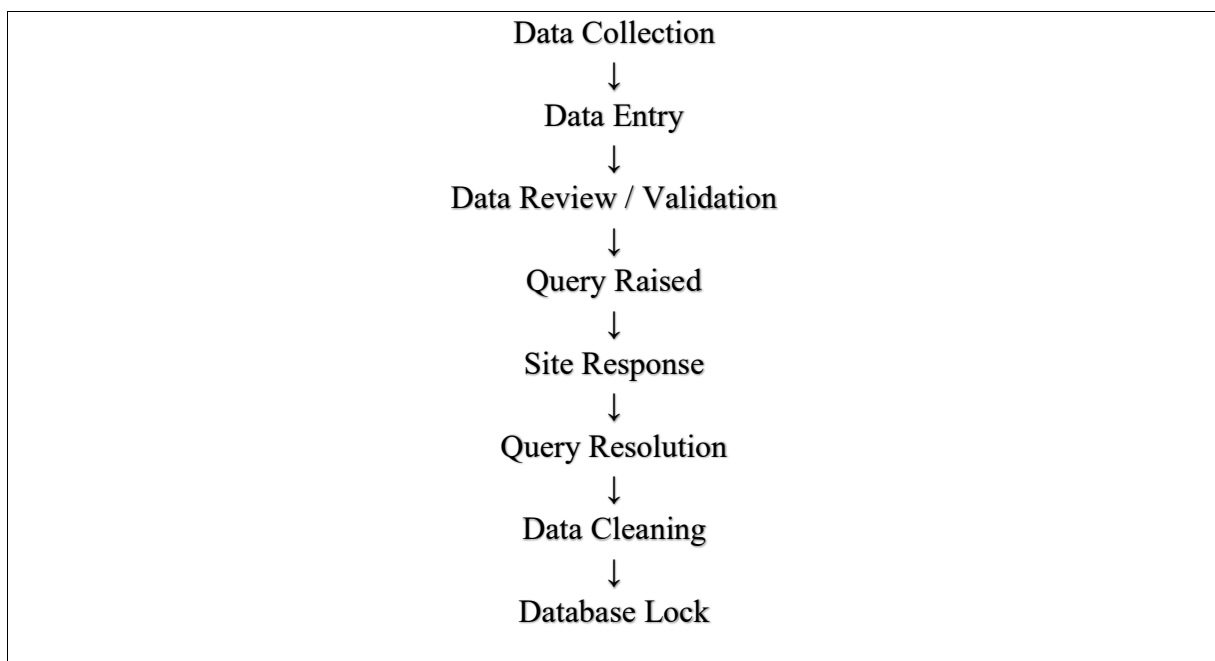
Validation system auto edit checks are generally triggered at the time of entry and include range checks, mandatory fields, consistency, and visit-day windows. Queries (Q) are raised in the EDC system to clarify or explain data discrepancies, mismatched dates, and missing visits. Auto-validation checks are system edit checks incorporated into the EDC system to automatically verify the data upon saving the form.

5) QUERY MANAGEMENT

The standard procedure when a query is raised for data involves isolating the disputed data, investigating the issue, and either correcting or validating the information. This process ensures data integrity and maintains trust between data consumers and providers.

After data entry and validation, if any discrepancy is found in the data, a query is raised. The query is sent to the concerned site for clarification or correction. After receiving the response, the data is updated, verified, and then moved to the next stage of cleaning and analysis.

FLOWCHART :- WHEN QUERY RAISED



6) MEDICAL CODING

Medical coding is the process of converting verbatim text into standardized, universal medical codes. This process enables the use of common languages to describe clinical observations and procedures worldwide. The standard dictionaries (MedDRA) are used for coding adverse events (AE) and medical history terms.

7) DATA RECONCILIATION

Data reconciliation is a data cleaning process where two different data sets that contain some similar information are compared, reviewed, and assessed to ensure that they are consistent and accurate. The two data sets are examined manually and validated against each other to identify differences, which are then corrected.

The objective of data reconciliation is to ensure that all essential safety and efficacy data match precisely before database lock.

8) DATABASE LOCK

Database lock (DBL) is a crucial phase and marks the end of the clinical trial. It is the formal closure of the database after all the data have been captured, queries answered, data cleaned, and reconciled, and the final data undergo validation and are ready for statistical analysis and regulatory submission.

Database lock criteria include:

- All subjects and visits entered
- All queries closed or commented
- Coding completed and reconciled

9) CONCLUSION

Clinical trial data are entered into the EDC through the forms where they are automatically checked, cleaned, coded, reconciled, and locked.

- FLOW CHART: COMPLETE CLINICAL TRIAL DATA LIFECYCLE

PRE DATA COLLECTION



DATA COLLECTION



ENTRY INTO EDC



DATA CLEANING AND AUTO VALIDATION CHECK



QUERY MANAGEMENT



MEDICAL CODING DATA RECONCILIATION



DATA RECONCILIATION



DATABASE LOCK